

Analysis

Introduction

Social Media has become a primary avenue for many consumers to consume various information such as news, health advice, entertainment, and many others. With many AI capabilities, Social Media has become more influential in targeting individuals with the information of their interest. To the most extent, information related to entertainment, advertisements might be okay. However, when individuals are exposed with health related information, not only experts in this field are sharing their perspective, we are also seeing many influencers sharing their perspective based on their experience, their observations, etc.

While often influencers generate higher volume of engagement and visibility, experts are mostly considered as credible due to their knowledge, education, and expertise. This raises an important question, does the source of information itself shape how the information is perceived by users and how do they act on these advices?

In real-world contexts, it's challenging to determine the effect of the source because posts from experts and influencers vary across multiple dimensions, including varying audiences, and presentation styles. Consequently, observational comparisons cannot accurately assess whether differences in engagement or credibility are genuinely influenced by the source or if they are due to these confounding factors.

To address this challenge, this experiment utilizes a randomized controlled experiment that keeps the content of a nutrition message consistent while varying only the source label (either expert or influencer). By randomly assigning participants to different conditions, the design isolates the causal impact of source cues on key outcomes, such as perceived credibility, trust, and intention to follow the advice. This method allows for a clear examination of whether labeling information as originating from an expert rather than an influencer significantly changes how individuals evaluate and respond to nutritional content.

Research Question

Are expert recommendations perceived as more credible than non-expert recommendations in the context of social media nutritional posts or recommendations?

Hypotheses

Null Hypothesis (H0):

There is no difference in credibility, trust, or behavioral intention between participants exposed to expert and influencer source cues. $H_0 : E[Y|Expert] = E[Y|Influencer]$

Alternative Hypothesis (H1):

Participants exposed to the expert source cue will report higher credibility, trust, and behavioral intention than those exposed to the influencer source cue. $H_a : E[Y|Expert] > E[Y|Influencer]$

where Y represents outcome such as perceived credibility, trust, or behavioral intention

PROCESS DATA

```
file <- 'SocialMediaViews_April 8, 2026_21.02.csv'
d_raw <- fread(file = here("final_project", "analysis", file), na.strings = c("", "NA"))

d <- d_raw[-(1:2)]
d[d == -99] <- NA
d[, complete := ifelse(is.na(complete), 0, complete)]

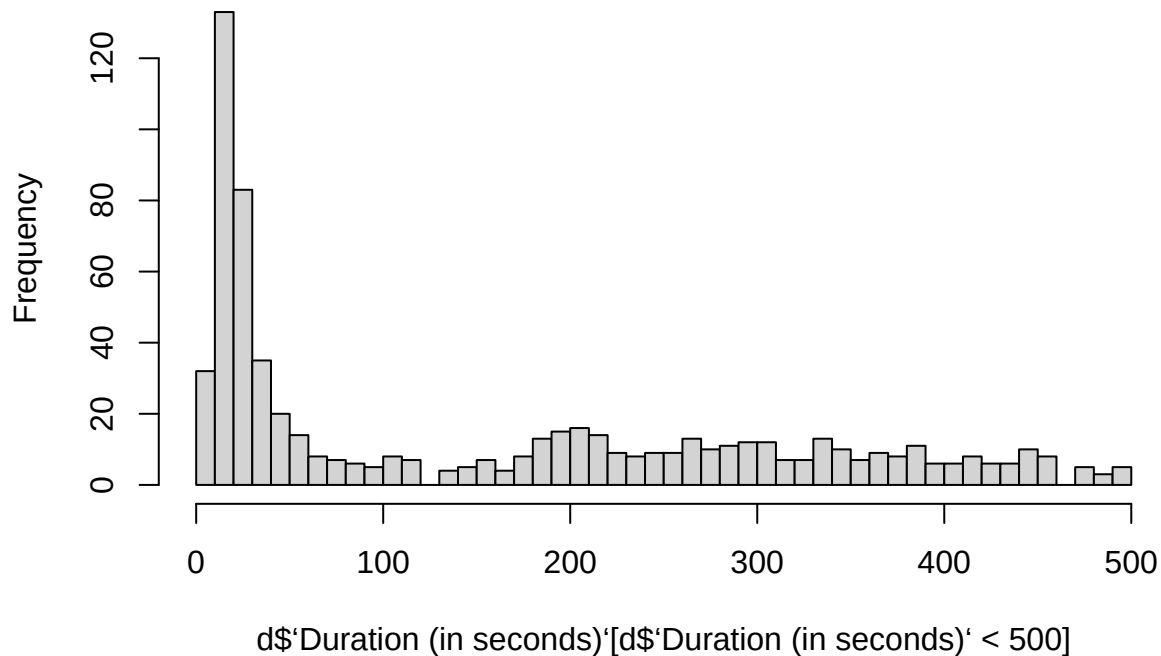
keep <- c(
  "ResponseId", "Finished", "Duration (in seconds)", "Status", "SCO",
  "gender", "age", "race", "hispanic_origin", "education",
  "country_origin", "region", "politics",
  grep("expert_trust", names(d), value = TRUE),
  grep("health_engagement", names(d), value = TRUE),
  "info_source",
  grep("^attention", names(d), value = TRUE),
  "Q_RecaptchaScore", "Q_DuplicateRespondent", "complete",
  grep("accurate|authentic|believable", names(d), value = TRUE),
  "condition_sequence", "order_pattern",
  "post_meat_source", "post_oil_source",
  "IPAddress"
)

d <- d[, ..keep]

char_cols <- c("ResponseId", "info_source", "condition_sequence", "order_pattern",
              "post_meat_source", "post_oil_source", "IPAddress")
num_cols <- setdiff(names(d), char_cols)
d[, (num_cols) := lapply(.SD, function(x) suppressWarnings(as.numeric(x))), .SDcols = num_cols]

d$Q_DuplicateRespondent[is.na(d$Q_DuplicateRespondent)] <- 0
hist(d$`Duration (in seconds)`[d$`Duration (in seconds)` < 500], breaks = 50)
```

Histogram of d\$'Duration (in seconds)'[d\$'Duration (in seconds)' < 500]



```
d[, keep_flag := as.integer(
  (!is.na(Q_RecaptchaScore) & Q_RecaptchaScore >= 0.5) &
  (!is.na(complete) & complete == 1) &
  (!is.na(SC0) & SC0 <= 1) &
  (is.na(Q_DuplicateRespondent) | Q_DuplicateRespondent != TRUE) &
  (!is.na(`Duration (in seconds)`) & `Duration (in seconds)` >= 120)
)]
sum(d$keep_flag)

## [1] 258

d[, expert_trust_index := rowMeans(.SD, na.rm = TRUE),
  .SDcols = c("1_expert_trust", "2_expert_trust", "3_expert_trust", "4_expert_trust", "5_expert_trust")]

d[, health_engagement_index := rowMeans(.SD, na.rm = TRUE),
  .SDcols = c("1_health_engagement", "2_health_engagement", "3_health_engagement",
    "4_health_engagement", "5_health_engagement")]

d[, meat_exp_cred := rowMeans(.SD, na.rm = TRUE),
  .SDcols = c("meat_exp_accurate", "meat_exp_authentic", "meat_exp_believable")]
d[, meat_infl_cred := rowMeans(.SD, na.rm = TRUE),
  .SDcols = c("meat_infl_accurate", "meat_infl_authentic", "meat_infl_believable")]
d[, meat_cred := fifelse(post_meat_source == "expert", meat_exp_cred, meat_infl_cred)]

d[, oil_exp_cred := rowMeans(.SD, na.rm = TRUE),
  .SDcols = c("oil_exp_accurate", "oil_exp_authentic", "oil_exp_believable")]
d[, oil_infl_cred := rowMeans(.SD, na.rm = TRUE),
  .SDcols = c("oil_infl_accurate", "post_oil_authentic", "post_oil_believable")]
d[, oil_cred := fifelse(post_oil_source == "expert", oil_exp_cred, oil_infl_cred)]
```

```
d[, age := factor(age, levels = 1:7,
  labels = c("Under18", "18-24", "25-34", "35-44", "45-54", "55-64", "65+"))]
d[, age_group := ifelse(age %in% c("18-24", "25-34"), "18-34",
  fifelse(age %in% c("35-44", "45-54"), "35-54", "55+"))]
d[, age_group := factor(age_group)]

saveRDS(d, here("clean_data_zr.rds"))
fwrite(d, here("clean_data_zr.csv"))
```

LOAD DATA

```
d_rds <- readRDS(file = here("clean_data_zr.rds"))
d <- d_rds[keep_flag == 1]
d <- d[SCO == 0]
```

DESCRIPTIVE STATISTICS

```
vars <- c("expert_trust_index", "health_engagement_index",
  "meat_exp_cred", "meat_infl_cred", "oil_exp_cred", "oil_infl_cred",
  "oil_cred", "meat_cred")

stargazer(as.data.frame(d[, ..vars]), type = "latex", header = FALSE,
  title = "Descriptive Statistics for Pilot Variables",
  summary.stat = c("mean", "sd", "min", "max"))
```

Table 1: Descriptive Statistics for Pilot Variables

Statistic	Mean	St. Dev.	Min	Max
expert_trust_index	4.995	0.826	1.000	7.000
health_engagement_index	5.343	1.029	1.000	7.000
meat_exp_cred	4.580	1.635	1.000	7.000
meat_infl_cred	4.341	1.689	1.000	7.000
oil_exp_cred	5.097	1.412	1.000	7.000
oil_infl_cred	4.700	1.320	1.000	7.000
oil_cred	4.884	1.374	1.000	7.000
meat_cred	4.470	1.660	1.000	7.000

```
lapply(d[, ..vars], function(x) sum(is.na(x)))
```

```
$expert_trust_index [1] 0
```

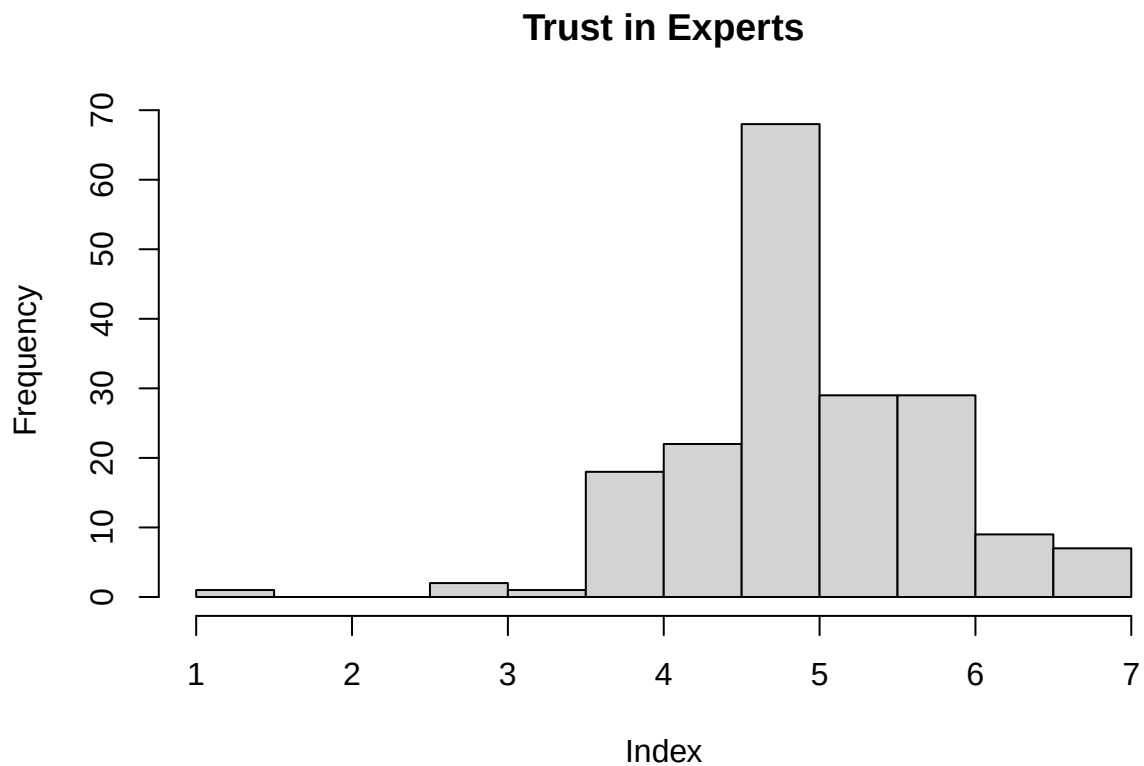
```

$health_engagement_index [1] 0
$meat_exp_cred [1] 86
$meat_infl_cred [1] 100
$oil_exp_cred [1] 100
$oil_infl_cred [1] 86
$oil_cred [1] 0
$meat_cred [1] 0

```

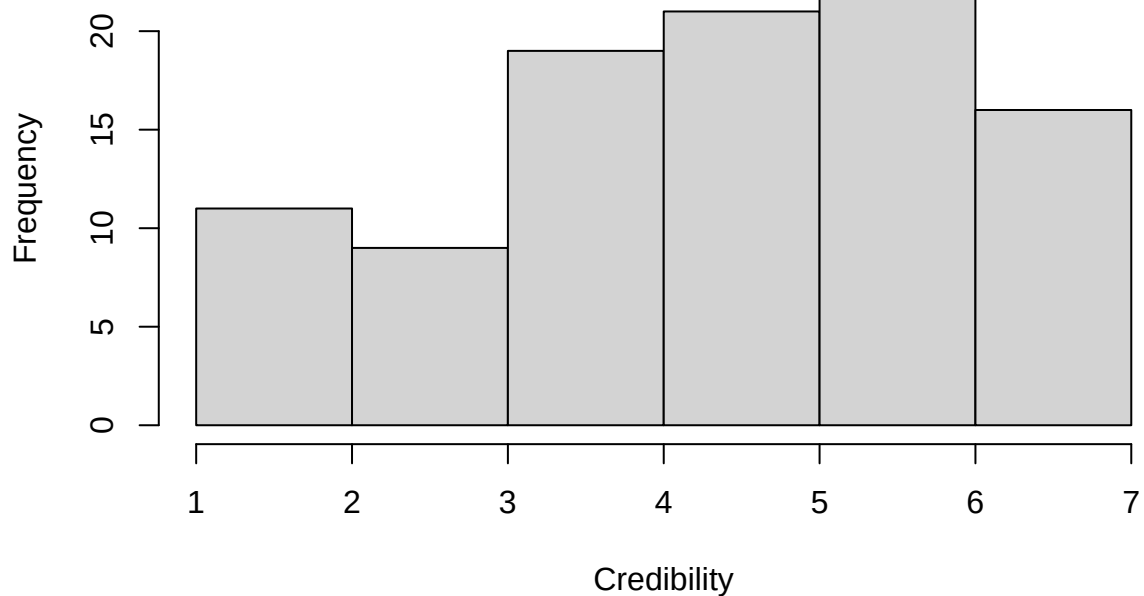
HISTOGRAMS

```
hist(d$expert_trust_index, main = "Trust in Experts", xlab = "Index")
```



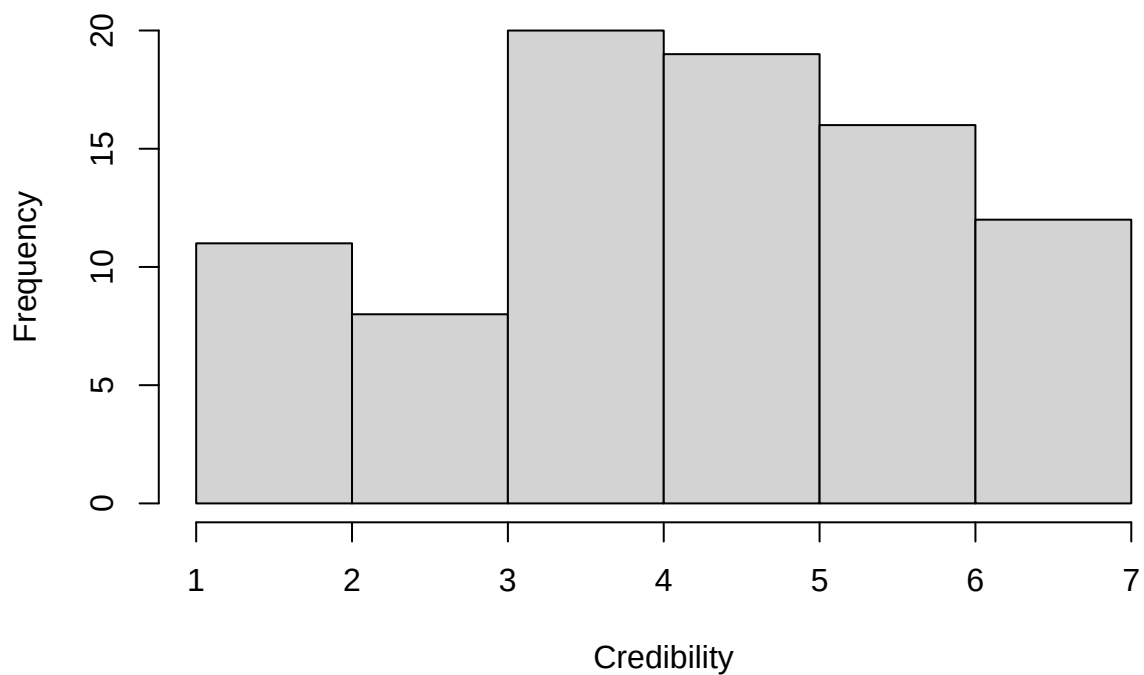
```
hist(d$meat_exp_cred, main = "Meat Expert Credibility", xlab = "Credibility")
```

Meat Expert Credibility



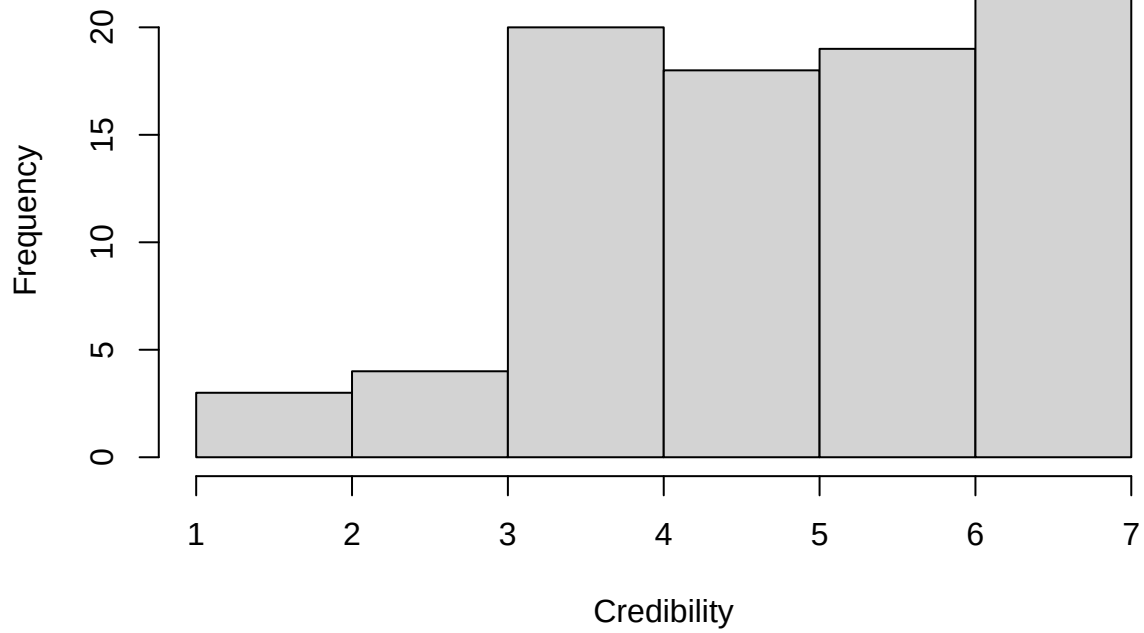
```
hist(d$meat_infl_cred,      main = "Meat Influencer Credibility", xlab = "Credibility")
```

Meat Influencer Credibility



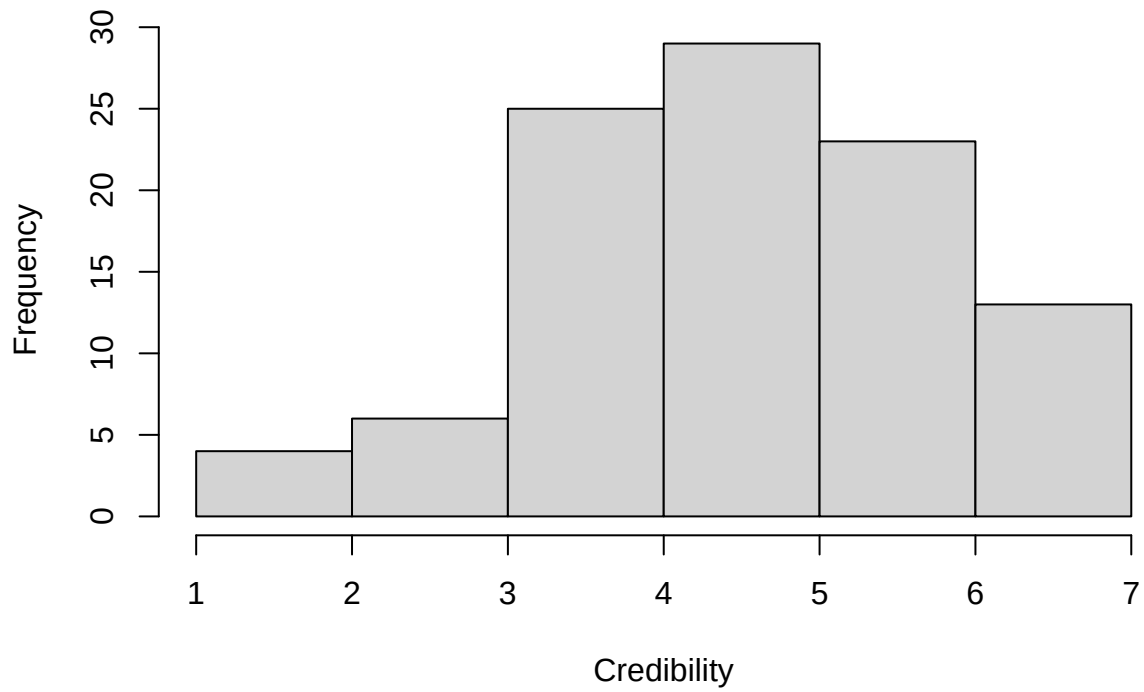
```
hist(d$oil_exp_cred,      main = "Oil Expert Credibility",      xlab = "Credibility")
```

Oil Expert Credibility



```
hist(d$oil_infl_cred,      main = "Oil Influencer Credibility", xlab = "Credibility")
```

Oil Influencer Credibility



PAIRED T-TESTS

```
t_meat <- t.test(d$meat_exp_cred, d$meat_infl_cred, paired = FALSE)
print(t_meat)

##
## Welch Two Sample t-test
##
## data: d$meat_exp_cred and d$meat_infl_cred
## t = 0.97625, df = 177.97, p-value = 0.3303
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -0.2440239 0.7218534
## sample estimates:
## mean of x mean of y
## 4.580000 4.341085

t_oil <- t.test(d$oil_exp_cred, d$oil_infl_cred, paired = FALSE)
print(t_oil)

##
## Welch Two Sample t-test
##
## data: d$oil_exp_cred and d$oil_infl_cred
## t = 1.9693, df = 175.61, p-value = 0.05049
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -0.0008648816 0.7946633312
## sample estimates:
## mean of x mean of y
## 5.096899 4.700000
```

REGRESSION ANALYSIS

```
model_oil_1 <- lm(oil_cred ~ post_oil_source, data = d)
coeftest(model_oil_1, vcov = vcovHC(model_oil_1, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.09690    0.15222  33.4841  <2e-16 ***
## post_oil_sourceinfluencer -0.39690    0.20153  -1.9694   0.0504 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_oil_2 <- lm(oil_cred ~ post_oil_source + expert_trust_index + health_engagement_index, data = d)
coeftest(model_oil_2, vcov = vcovHC(model_oil_2, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.87125    0.64170  4.4744 1.347e-05 ***
## post_oil_sourceinfluencer -0.32199    0.19597 -1.6431  0.10210
## expert_trust_index      0.21998    0.12642  1.7401  0.08354 .
## health_engagement_index  0.20338    0.11926  1.7053  0.08985 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_oil_3 <- lm(oil_cred ~ post_oil_source + expert_trust_index + health_engagement_index + age + gender, data = d)
coeftest(model_oil_3, vcov = vcovHC(model_oil_3, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.6172095    0.7746801  3.3784 0.0008986 ***
## post_oil_sourceinfluencer -0.3271768    0.1932008 -1.6935 0.0921481 .
## expert_trust_index      0.1474594    0.1268857  1.1621 0.2467597
## health_engagement_index  0.2010336    0.1133592  1.7734 0.0778982 .
## age18-24           0.9913113    0.4127061  2.4020 0.0173533 *
## age25-34           0.4037320    0.4241888  0.9518 0.3425242
## age35-44           0.7002998    0.3805825  1.8401 0.0674513 .
## age45-54           0.4027709    0.3934936  1.0236 0.3074481
## age55-64           0.5019916    0.3799992  1.3210 0.1882150
## gender             0.0716416    0.2048275  0.3498 0.7269354
## education          -0.0050693    0.0854028 -0.0594 0.9527348
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_oil_4 <- lm(oil_cred ~ post_oil_source + health_engagement_index + post_oil_source * health_engagement_index, data = d)
coeftest(model_oil_4, vcov = vcovHC(model_oil_4, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.40394    0.86340  2.7843 0.0064811 **
## post_oil_sourceinfluencer  1.45569    1.10736  1.3146 0.1891111
## health_engagement_index  0.48963    0.16370  2.9911 0.0026111
## post_oil_sourceinfluencer:health_engagement_index -0.32827    0.21052 -1.5593 0.120662
## Pr(>|t|)
## (Intercept)      0.005932 **
## post_oil_sourceinfluencer  0.190312
## health_engagement_index  0.003165 **
## post_oil_sourceinfluencer:health_engagement_index 0.120662
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

model_meat_1 <- lm(meat_cred ~ post_meat_source, data = d)
coeftest(model_meat_1, vcov = vcovHC(model_meat_1, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.58000    0.16354 28.0051  <2e-16 ***
## post_meat_sourceinfluencer -0.23891    0.24471 -0.9763   0.3302
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_meat_2 <- lm(meat_cred ~ post_meat_source + expert_trust_index + health_engagement_index, data = d)
coeftest(model_meat_2, vcov = vcovHC(model_meat_2, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.56522    0.79685  1.9643 0.051022 .
## post_meat_sourceinfluencer -0.36809    0.23170 -1.5886 0.113881
## expert_trust_index      0.19195    0.14368  1.3360 0.183223
## health_engagement_index   0.39599    0.14057  2.8170 0.005383 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_meat_3 <- lm(meat_cred ~ post_meat_source + expert_trust_index + health_engagement_index + age + gender + education)
coeftest(model_meat_3, vcov = vcovHC(model_meat_3, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.618517    1.021066  1.5851 0.114743
## post_meat_sourceinfluencer -0.370590    0.230842 -1.6054 0.110213
## expert_trust_index      0.191311    0.146405  1.3067 0.193018
## health_engagement_index   0.424813    0.135137  3.1436 0.001961 **
## age18-24           0.153503    0.440827  0.3482 0.728097
## age25-34          -0.763773    0.477254 -1.6003 0.111325
## age35-44           0.248724    0.408073  0.6095 0.542978
## age45-54           0.100206    0.444350  0.2255 0.821845
## age55-64           0.482149    0.428933  1.1241 0.262524
## gender             -0.021169    0.239065 -0.0886 0.929540
## education          -0.084579    0.112526 -0.7516 0.453277
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_meat_4 <- lm(meat_cred ~ post_meat_source + health_engagement_index + post_meat_source * health_engagement_index)
coeftest(model_meat_4, vcov = vcovHC(model_meat_4, type = "HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value
## (Intercept)      2.85170    0.87978  3.2414
## post_meat_sourceinfluencer -2.33117    1.38148 -1.6874

```

```
## health_engagement_index          0.33185    0.16104    2.0606
## post_meat_sourceinfluencer:health_engagement_index 0.36279    0.25317    1.4330
##                                Pr(>|t|)
## (Intercept)                    0.001415 **
## post_meat_sourceinfluencer      0.093231 .
## health_engagement_index        0.040760 *
## post_meat_sourceinfluencer:health_engagement_index 0.153571
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

stargazer(model_meat_1, model_meat_2, model_meat_3, model_meat_4,
  dep.var.labels = "Meat Credibility",
  title          = "Meat Regressions",
  type           = "latex",
  covariate.labels = c("Influencer", "Expert Trust Index", "Health Engagement Index",
    "Age 18-24", "Age 25-34", "Age 35-44", "Age 45-54", "Age 55-64",
    "Gender", "Education", "Influencer*Health Engagement"),
  column.sep.width = "2pt",
  omit.stat        = "F",
  header           = FALSE)
```

Table 2: Meat Regressions

	<i>Dependent variable:</i>			
	Meat Credibility			
	(1)	(2)	(3)	(4)
Influencer	−0.239 (0.244)	−0.368 (0.237)	−0.371 (0.235)	−2.331* (1.311)
Expert Trust Index		0.192 (0.157)	0.191 (0.160)	
Health Engagement Index		0.396*** (0.127)	0.425*** (0.127)	0.332** (0.143)
Age 18-24			0.154 (0.447)	
Age 25-34			−0.764 (0.469)	
Age 35-44			0.249 (0.468)	
Age 45-54			0.100 (0.442)	
Age 55-64			0.482 (0.436)	
Gender			−0.021 (0.240)	
Education			−0.085 (0.108)	
Influencer*Health Engagement				0.363 (0.239)
Constant	4.580*** (0.166)	1.565* (0.794)	1.619* (0.938)	2.852*** (0.764)
Observations	186	186	186	186
R ²	0.005	0.093	0.148	0.097
Adjusted R ²	−0.0002	0.078	0.099	0.082
Residual Std. Error	1.660 (df = 184)	1.594 (df = 182)	1.575 (df = 175)	1.590 (df = 182)

Note:

*p<0.1; **p<0.05; ***p<0.01

```
stargazer(model_oil_1, model_oil_2, model_oil_3, model_oil_4,
  title       = "Oil Regressions",
  type        = "latex",
  dep.var.labels = "Oil Credibility",
  covariate.labels = c("Influencer", "Expert Trust Index", "Health Engagement Index",
    "Age 18-24", "Age 25-34", "Age 35-44", "Age 45-54", "Age 55-64",
    "Gender", "Education", "Influencer*Health Engagement"),
  column.sep.width = "2pt",
  omit.stat        = "F",
  header           = FALSE)
```

=====

WITHIN-SUBJECT MODELS

=====

```
d_long <- d %>%
  rename(
    oil_infl_authentic = post_oil_authentic,
    oil_infl_believable = post_oil_believable,
    dmy1_slp_attention = attention1,
    dmy2_wt_attention = attention2,
    oil_infl_attention = attention_oil_inf,
    oil_exp_attention = attention_oil_exp,
    meat_infl_attention = attention_meat_infl,
    meat_exp_attention = attention_meat_exp
  ) %>%
  pivot_longer(
    cols = c(
      meat_exp_accurate, meat_exp_authentic, meat_exp_believable, meat_infl_accurate,
      meat_exp_attention, meat_infl_authentic, meat_infl_believable, meat_exp_cred,
      meat_infl_cred, meat_infl_attention,
      oil_exp_accurate, oil_exp_authentic, oil_exp_believable, oil_exp_cred,
      oil_exp_attention, oil_infl_accurate, oil_infl_cred, oil_infl_authentic,
      oil_infl_believable, oil_infl_attention,
      dmy1_slp_accurate, dmy1_slp_authentic, dmy1_slp_believable, dmy1_slp_attention,
      dmy2_wt_accurate, dmy2_wt_authentic, dmy2_wt_believable, dmy2_wt_attention
    ),
    names_to = "variable",
    values_to = "value"
  ) %>%
  mutate(
    question = case_when(
      grepl("^meat", variable) ~ "meat",
      grepl("^oil", variable) ~ "oil",
      grepl("^dmy1", variable) ~ "dmy1",
      grepl("^dmy2", variable) ~ "dmy2"
    ),
    inf_expert = case_when(
      grepl("exp", variable) ~ "expert",
```

Table 3: Oil Regressions

	<i>Dependent variable:</i>			
	Oil Credibility			
	(1)	(2)	(3)	(4)
Influencer	−0.397** (0.201)	−0.322 (0.198)	−0.327 (0.198)	1.456 (1.097)
Expert Trust Index		0.220* (0.131)	0.147 (0.135)	
Health Engagement Index		0.203* (0.106)	0.201* (0.107)	0.490*** (0.160)
Age 18-24			0.991*** (0.376)	
Age 25-34			0.404 (0.395)	
Age 35-44			0.700* (0.394)	
Age 45-54			0.403 (0.371)	
Age 55-64			0.502 (0.367)	
Gender			0.072 (0.202)	
Education			−0.005 (0.091)	
Influencer*Health Engagement				−0.328 (0.200)
Constant	5.097*** (0.147)	2.871*** (0.684)	2.617*** (0.813)	2.404*** (0.892)
Observations	186	186	186	186
R ²	0.021	0.078	0.121	0.077
Adjusted R ²	0.016	0.063	0.071	0.062
Residual Std. Error	1.364 (df = 184)	1.330 (df = 182)	1.325 (df = 175)	1.331 (df = 182)

Note:

*p<0.1; **p<0.05; ***p<0.01

```

    grepl("infl", variable) ~ "influencer"
  ),
  metric = case_when(
    grepl("accurate", variable) ~ "accurate",
    grepl("authentic", variable) ~ "authentic",
    grepl("believable", variable) ~ "believable",
    grepl("cred", variable) ~ "cred",
    grepl("attention", variable) ~ "attention"
  )
) %>%
group_by(across(-c(variable, value))) %>%
summarise(value = mean(value, na.rm = TRUE), .groups = "drop") %>%
pivot_wider(names_from = metric, values_from = value) %>%
mutate(
  inf_expert = ifelse(is.na(inf_expert), "", inf_expert),
  question_inf_expert = trimws(paste0(question, inf_expert))
) %>%
mutate(
  position = case_when(
    order_pattern == "tfft" & question == "meat" ~ 1,
    order_pattern == "tfft" & question == "dmy1" ~ 2,
    order_pattern == "tfft" & question == "dmy2" ~ 3,
    order_pattern == "tfft" & question == "oil" ~ 4,
    order_pattern == "ftft" & question == "dmy1" ~ 1,
    order_pattern == "ftft" & question == "meat" ~ 2,
    order_pattern == "ftft" & question == "dmy2" ~ 3,
    order_pattern == "ftft" & question == "oil" ~ 4
  )
) %>%
select(ResponseId, condition_sequence, order_pattern, question, inf_expert,
  question_inf_expert, position, accurate, authentic, believable, cred,
  attention, everything()) %>%
filter(!(condition_sequence == "influencer_meat_expert_oil" &
  question_inf_expert %in% c("meatexpert", "oilinfluencer"))) %>%
filter(!(condition_sequence == "expert_meat_influencer_oil" &
  question_inf_expert %in% c("meatinfluencer", "oilexpert")))

d_long_within <- filter(d_long, inf_expert %in% c("expert", "influencer"))
stopifnot(nrow(d_long_within) == 2 * nrow(d))

model_within_1 <- lm(cred ~ inf_expert + factor(ResponseId), data = d_long_within)

res_1 <- tidy(coeftest(model_within_1, vcov = vcovCL(model_within_1, cluster = ~ResponseId)))
res_1[grepl("inf_expert", res_1$term), ]

```

A tibble: 1 x 5

```
term estimate std.error statistic p.value 1 inf_expertinfluencer -0.285 0.170 -1.68 0.0947
```

```
se_within_1 <- sqrt(diag(vcovCL(model_within_1, cluster = ~ResponseId)))
```

```
stargazer(model_within_1,
  se = list(se_within_1),
```

```

title           = "Within-Subjects Regression (Respondent Fixed Effects)",
dep.var.labels  = "Perceived Credibility",
covariate.labels = c("Influencer (vs Expert)"),
omit           = "ResponseId",
omit.labels     = "Respondent FE",
add.lines       = list(c("Respondent FE", "Yes")),
type           = "latex",
header          = FALSE,
omit.stat       = "f",
column.sep.width = "2pt")

```

Table 4: Within-Subjects Regression (Respondent Fixed Effects)

	<i>Dependent variable:</i>
	Perceived Credibility
Influencer (vs Expert)	−0.285* (0.170)
Constant	5.142*** (0.085)
Respondent FE	Yes
Respondent FE	Yes
Observations	372
R ²	0.718
Adjusted R ²	0.434
Residual Std. Error	1.155 (df = 185)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

POOLED WITHIN-SUBJECT MODELS

```
d_pooled <- d %>%
  pivot_longer(
    cols      = c(meat_cred, oil_cred),
    names_to  = "claim",
    values_to = "cred"
  ) %>%
  mutate(
    claim      = gsub("_cred", "", claim),
    post_source = ifelse(claim == "meat", post_meat_source, post_oil_source),
    is_expert  = as.integer(post_source == "expert")
  ) %>%
  as.data.frame()

model_pool_1 <- lm(cred ~ is_expert + factor(ResponseId), data = d_pooled)
model_pool_2 <- lm(cred ~ is_expert + expert_trust_index + health_engagement_index +
  factor(ResponseId), data = d_pooled)
model_pool_3 <- lm(cred ~ is_expert + expert_trust_index + health_engagement_index +
  age + gender + education + factor(ResponseId), data = d_pooled)
model_pool_4 <- lm(cred ~ is_expert * health_engagement_index +
  factor(ResponseId), data = d_pooled)

se_pool_1 <- sqrt(diag(vcovCL(model_pool_1, cluster = ~ResponseId)))
se_pool_2 <- sqrt(diag(vcovCL(model_pool_2, cluster = ~ResponseId)))
se_pool_3 <- sqrt(diag(vcovCL(model_pool_3, cluster = ~ResponseId)))
se_pool_4 <- sqrt(diag(vcovCL(model_pool_4, cluster = ~ResponseId)))

stargazer(
  model_pool_1, model_pool_2, model_pool_3, model_pool_4,
  se          = list(se_pool_1, se_pool_2, se_pool_3, se_pool_4),
  title       = "Pooled Within-Subject Regressions (Meat \\& Oil)",
  dep.var.labels = "Perceived Credibility",
  covariate.labels = c(
    "Expert (vs Influencer)",
    "Expert Trust Index",
    "Health Engagement Index",
    "Age 18-24", "Age 25-34", "Age 35-44", "Age 45-54", "Age 55-64",
    "Gender", "Education",
    "Expert $\\times$ Health Engagement"
  ),
  omit        = "ResponseId",
  omit.labels = "Respondent FE",
  add.lines   = list(c("Respondent FE", "Yes", "Yes", "Yes", "Yes")),
  type        = "latex",
  column.sep.width = "2pt",
  omit.stat    = "f",
  header      = FALSE)
```

Table 5: Pooled Within-Subject Regressions (Meat & Oil)

	<i>Dependent variable:</i>			
	Perceived Credibility			
	(1)	(2)	(3)	(4)
Expert (vs Influencer)	0.285* (0.170)	0.285* (0.170)	0.285* (0.170)	-0.223 (0.775)
Expert Trust Index		-0.278*** (0.000)	-3.333*** (0.000)	
Health Engagement Index		0.889*** (0.000)	-0.833*** (0.000)	0.619*** (0.073)
Age 18-24			8.333*** (0.000)	
Age 25-34			3.000*** (0.000)	
Age 35-44			4.333*** (0.000)	
Age 45-54			1.500*** (0.000)	
Age 55-64			-0.333*** (0.000)	
Gender			0.167*** (0.000)	
Education			1.000*** (0.000)	
Expert $\times$ Health Engagement				0.095 (0.145)
Constant	4.858*** (0.085)	1.213*** (0.085)	19.024*** (0.085)	1.378*** (0.388)
Respondent FE	Yes	Yes	Yes	Yes
Respondent FE	Yes	Yes	Yes	Yes
Observations	372	372	372	372
R ²	0.718	0.718	0.718	0.719
Adjusted R ²	0.434	0.434	0.434	0.433
Residual Std. Error	1.155 (df = 185)	1.155 (df = 185)	1.155 (df = 185)	1.156 (df = 184)

Note:

*p<0.1; **p<0.05; ***p<0.01

```

d_pooled <- d %>%
  pivot_longer(
    cols      = c(meat_cred, oil_cred),
    names_to  = "claim",
    values_to = "cred"
  ) %>%
  mutate(
    claim      = gsub("_cred", "", claim),
    post_source = ifelse(claim == "meat", post_meat_source, post_oil_source),
    is_influencer = as.integer(post_source == "influencer")
  ) %>%
  as.data.frame()

p1 <- lm(cred ~ is_influencer + factor(ResponseId), data = d_pooled)
p2 <- lm(cred ~ is_influencer + expert_trust_index + health_engagement_index +
  factor(ResponseId), data = d_pooled)
p3 <- lm(cred ~ is_influencer + expert_trust_index + health_engagement_index +
  age + gender + education + factor(ResponseId), data = d_pooled)
p4 <- lm(cred ~ is_influencer * health_engagement_index +
  factor(ResponseId), data = d_pooled)

sp1 <- sqrt(diag(vcovCL(p1, cluster = ~ResponseId)))
sp2 <- sqrt(diag(vcovCL(p2, cluster = ~ResponseId)))
sp3 <- sqrt(diag(vcovCL(p3, cluster = ~ResponseId)))
sp4 <- sqrt(diag(vcovCL(p4, cluster = ~ResponseId)))

stargazer(p1, p2, p3, p4,
  se      = list(sp1, sp2, sp3, sp4),
  title    = "Pooled Within-Subject Regressions (Meat & Oil)",
  dep.var.labels = "Perceived Credibility",
  covariate.labels = c(
    "Influencer",
    "Expert Trust Index",
    "Health Engagement Index",
    "Age 18-24", "Age 25-34", "Age 35-44", "Age 45-54", "Age 55-64",
    "Gender", "Education",
    "Influencer  $\times$  Health Engagement"
  ),
  omit      = "ResponseId",
  omit.labels = "Respondent FE",
  add.lines = list(c("Respondent FE", "Yes", "Yes", "Yes", "Yes")),
  type      = "latex",
  column.sep.width = "2pt",
  omit.stat  = "f",
  header    = FALSE)

```

Table 6: Pooled Within-Subject Regressions (Meat & Oil)

	<i>Dependent variable:</i>			
	Perceived Credibility			
	(1)	(2)	(3)	(4)
Influencer	−0.285* (0.170)	−0.285* (0.170)	−0.285* (0.170)	0.223 (0.775)
Expert Trust Index		−0.278*** (0.000)	−3.333*** (0.000)	
Health Engagement Index		0.889*** (0.000)	−0.833*** (0.000)	0.714*** (0.073)
Age 18-24			8.333*** (0.000)	
Age 25-34			3.000*** (0.000)	
Age 35-44			4.333*** (0.000)	
Age 45-54			1.500*** (0.000)	
Age 55-64			−0.333*** (0.000)	
Gender			0.167*** (0.000)	
Education			1.000*** (0.000)	
Influencer × Health Engagement				−0.095 (0.145)
Constant	5.142*** (0.085)	1.498*** (0.085)	19.309*** (0.085)	1.155*** (0.388)
Respondent FE	Yes	Yes	Yes	Yes
Respondent FE	Yes	Yes	Yes	Yes
Observations	372	372	372	372
R ²	0.718	0.718	0.718	0.719
Adjusted R ²	0.434	0.434	0.434	0.433
Residual Std. Error	1.155 (df = 185)	1.155 (df = 185)	1.155 (df = 185)	1.156 (df = 184)

Note:

*p<0.1; **p<0.05; ***p<0.01

ATTRITION ANALYSIS

```
# Overall completion rate
completion_tbl <- data.frame(
  Metric = c("Total responses", "Completed", "Completion rate"),
  Value = c(
    nrow(d_rds),
    sum(d_rds$complete == 1, na.rm = TRUE),
    paste0(round(mean(d_rds$complete == 1, na.rm = TRUE) * 100, 1), "%")
  )
)
kable(completion_tbl, caption = "Overall Completion Rate", booktabs = TRUE)
```

Table 7: Overall Completion Rate

Metric	Value
Total responses	779
Completed	355
Completion rate	45.6%

```
# Sample retention after all filters
retention_tbl <- data.frame(
  Stage = c("All responses", "Passed all filters (keep_flag)", "Also passed attention (SC0 = 0)"),
  N = c(nrow(d_rds), sum(d_rds$keep_flag, na.rm = TRUE), nrow(d))
)
kable(retention_tbl, caption = "Sample Retention by Filtering Stage", booktabs = TRUE)
```

Table 8: Sample Retention by Filtering Stage

Stage	N
All responses	779
Passed all filters (keep_flag)	258
Also passed attention (SC0 = 0)	186

```
# Completion by meat source assignment
meat_tbl <- as.data.frame(
  prop.table(table(d_rds$post_meat_source, d_rds$complete), 1)
) %>%
  setNames(c("Meat Source", "Completed", "Proportion")) %>%
  mutate(Proportion = round(Proportion, 3))
kable(meat_tbl, caption = "Completion Rate by Meat Post Source Assignment", booktabs = TRUE)
```

Table 9: Completion Rate by Meat Post Source Assignment

Meat Source	Completed	Proportion
expert	0	0.062
influencer	0	0.069
expert	1	0.938
influencer	1	0.931

```
# Completion by oil source assignment
oil_tbl <- as.data.frame(
  prop.table(table(d_rds$post_oil_source, d_rds$complete), 1)
) %>%
  setNames(c("Oil Source", "Completed", "Proportion")) %>%
  mutate(Proportion = round(Proportion, 3))
kable(oil_tbl, caption = "Completion Rate by Oil Post Source Assignment", booktabs = TRUE)
```

Table 10: Completion Rate by Oil Post Source Assignment

Oil Source	Completed	Proportion
expert	0	0.069
influencer	0	0.062
expert	1	0.931
influencer	1	0.938

```
# Regress completion on treatment to test for differential attrition
model_attr_meat_1 <- lm(complete ~ post_meat_source, data = d_rds)
model_attr_meat_2 <- lm(complete ~ post_meat_source + age + gender + expert_trust_index, data = d_rds)

model_attr_oil_1 <- lm(complete ~ post_oil_source, data = d_rds)
model_attr_oil_2 <- lm(complete ~ post_oil_source + age + gender + expert_trust_index, data = d_rds)

se_attr_meat_1 <- sqrt(diag(vcovHC(model_attr_meat_1, type = "HC1")))
se_attr_meat_2 <- sqrt(diag(vcovHC(model_attr_meat_2, type = "HC1")))
se_attr_oil_1 <- sqrt(diag(vcovHC(model_attr_oil_1, type = "HC1")))
se_attr_oil_2 <- sqrt(diag(vcovHC(model_attr_oil_2, type = "HC1")))

m1 <- model_attr_meat_1
m2 <- model_attr_meat_2
m3 <- model_attr_oil_1
m4 <- model_attr_oil_2

se1 <- se_attr_meat_1
se2 <- se_attr_meat_2
se3 <- se_attr_oil_1
se4 <- se_attr_oil_2

stargazer(m1, m2, m3, m4,
  se = list(se1, se2, se3, se4),
  title = "Attrition Analysis: Treatment Predicting Survey Completion",
  dep.var.labels = c("Completed (Meat)", "Completed (Oil)"),
  covariate.labels = c("Influencer (Meat)", "Influencer (Oil)",
```

```

      "Age 18-24", "Age 25-34", "Age 35-44", "Age 45-54", "Age 55-64",
      "Gender", "Expert Trust Index"),
add.lines      = list(c("Demographic Controls", "No", "Yes", "No", "Yes")),
column.labels  = c("Meat Source", "Oil Source"),
column.separate = c(2, 2),
type           = "latex",
header         = FALSE,
omit.stat      = "f",
column.sep.width = "2pt",
notes          = "Robust (HC1) standard errors in parentheses. Dependent variable is binary (1 = completed survey.)"

m1 <- lm(complete ~ post_meat_source, data = d_rds)
m2 <- lm(complete ~ post_oil_source, data = d_rds)

se1 <- sqrt(diag(vcovHC(m1, type = "HC1")))
se2 <- sqrt(diag(vcovHC(m2, type = "HC1")))

stargazer(m1, m2,
  se      = list(se1, se2),
  title    = "Attrition Analysis: Treatment Predicting Survey Completion",
  dep.var.labels = "Completed (Binary)",
  column.labels  = c("Meat Source", "Oil Source"),
  covariate.labels = c("Influencer"),
  type          = "latex",
  header        = FALSE,
  omit.stat     = "f",
  column.sep.width = "2pt",
  notes         = "Robust (HC1) SEs. Dependent variable = 1 if completed survey.")

```

Table 11: Attrition Analysis: Treatment Predicting Survey Completion

	<i>Dependent variable:</i>			
	Meat Source		Completed (Meat)	Oil Source
	(1)	(2)	(3)	(4)
Influencer (Meat)	−0.007 (0.026)	−0.002 (0.026)		
Influencer (Oil)		−0.056 (0.049)		−0.056 (0.049)
Age 18-24		−0.053 (0.044)		−0.053 (0.044)
Age 25-34		0.036 (0.037)		0.036 (0.037)
Age 35-44		−0.002 (0.045)		−0.002 (0.045)
Age 45-54		−0.006 (0.042)		−0.006 (0.042)
Age 55-64		−0.053** (0.027)		−0.053** (0.027)
Gender		−0.003 (0.011)		−0.003 (0.011)
Expert Trust Index			0.007 (0.026)	0.002 (0.026)
Constant	0.937*** (0.018)	1.047*** (0.074)	0.931*** (0.019)	1.046*** (0.077)
Demographic Controls	No	Yes	No	Yes
Observations	380	380	380	380
R ²	0.0002	0.027	0.0002	0.027
Adjusted R ²	−0.002	0.006	−0.002	0.006
Residual Std. Error	0.249 (df = 378)	0.247 (df = 371)	0.249 (df = 378)	0.247 (df = 371)

Note:

*p<0.1; **p<0.05; ***p<0.01

Robust (HC1) standard errors in parentheses. Dependent variable is binary (1 = completed survey)

Table 12: Attrition Analysis: Treatment Predicting Survey Completion

	<i>Dependent variable:</i>	
	Meat Source (1)	Completed (Binary) Oil Source (2)
Influencer	−0.007 (0.026)	
post_oil_sourceinfluencer		0.007 (0.026)
Constant	0.937*** (0.018)	0.931*** (0.019)
Observations	380	380
R ²	0.0002	0.0002
Adjusted R ²	−0.002	−0.002
Residual Std. Error (df = 378)	0.249	0.249

Note:

*p<0.1; **p<0.05; ***p<0.01

Robust (HC1) SEs. Dependent variable = 1 if completed survey.

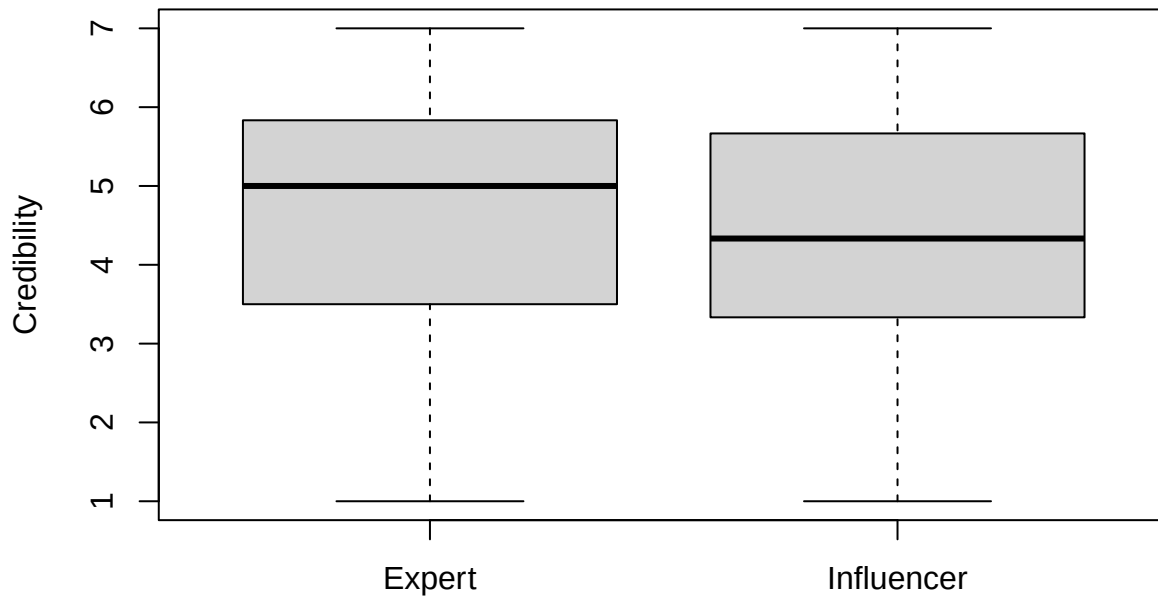
DATA VISUALIZATIONS

```

boxplot(d$meat_exp_cred, d$meat_infl_cred,
  names = c("Expert", "Influencer"),
  main = "Meat Credibility by Source", ylab = "Credibility")

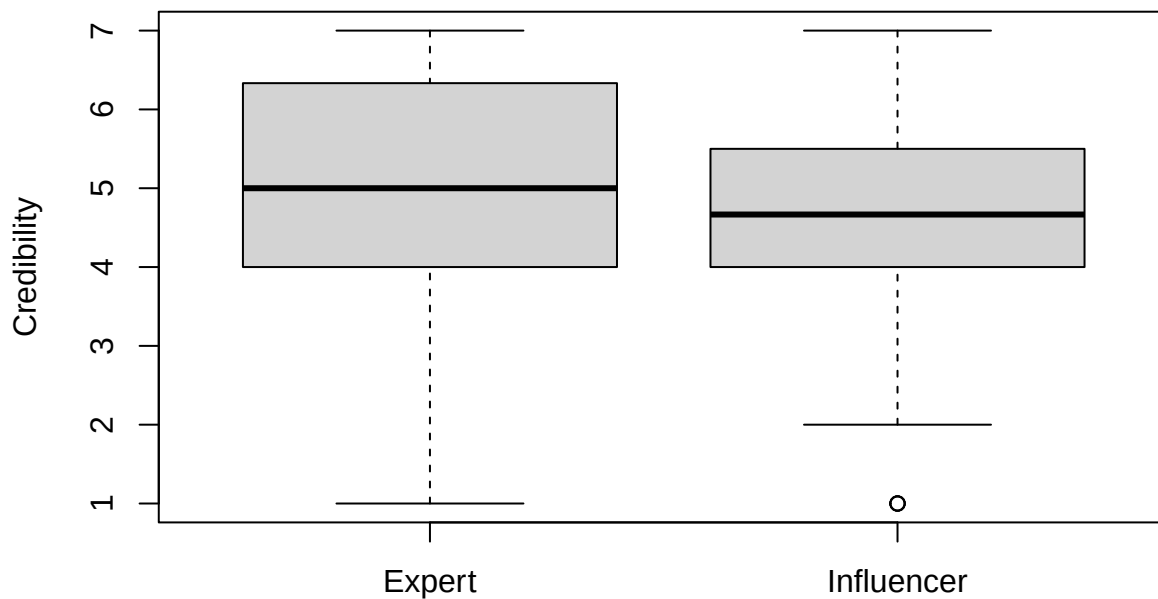
```

Meat Credibility by Source



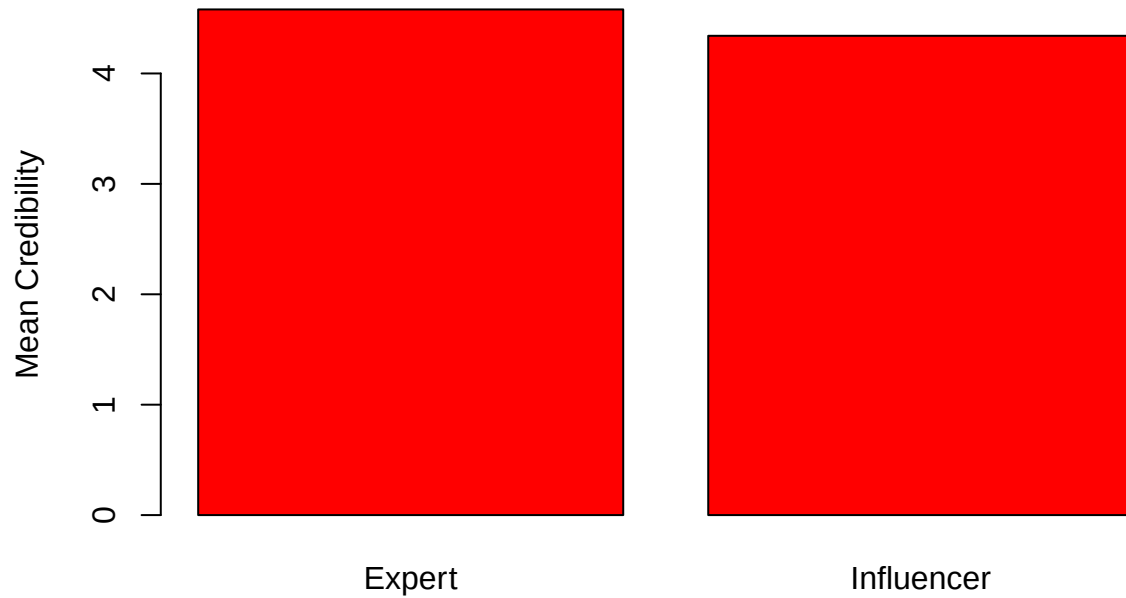
```
boxplot(d$oil_exp_cred, d$oil_infl_cred,
  names = c("Expert", "Influencer"),
  main = "Oil Credibility by Source", ylab = "Credibility")
```

Oil Credibility by Source



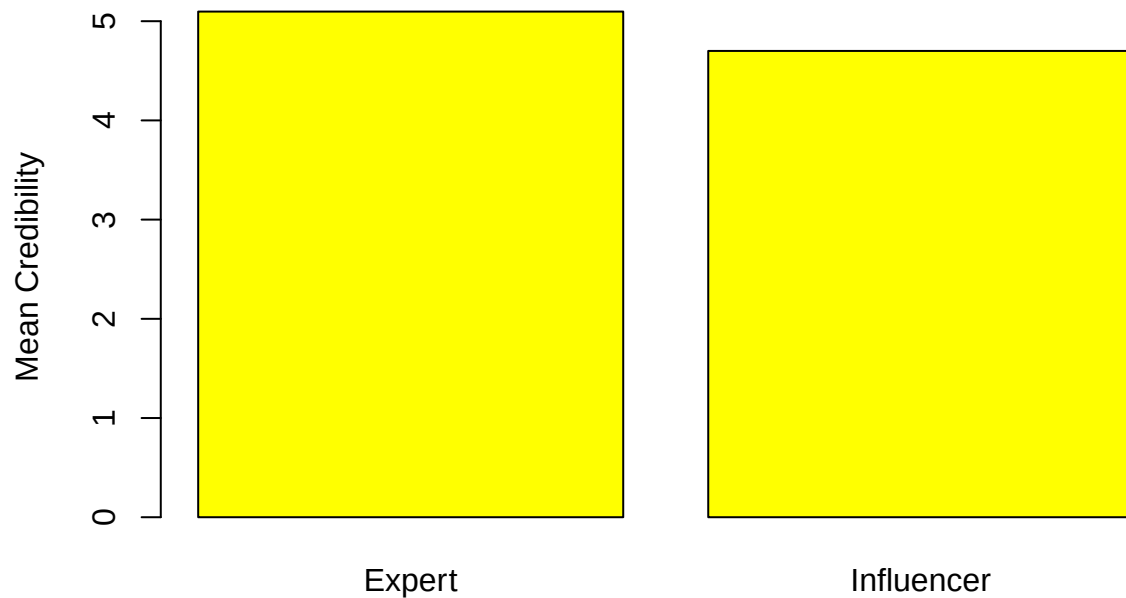
```
meat_means <- d[, .(Expert = mean(meat_exp_cred, na.rm = TRUE),
  Influencer = mean(meat_infl_cred, na.rm = TRUE))]
barplot(as.numeric(meat_means), names.arg = c("Expert", "Influencer"),
  main = "Average Meat Credibility", ylab = "Mean Credibility", col = "red")
```

Average Meat Credibility



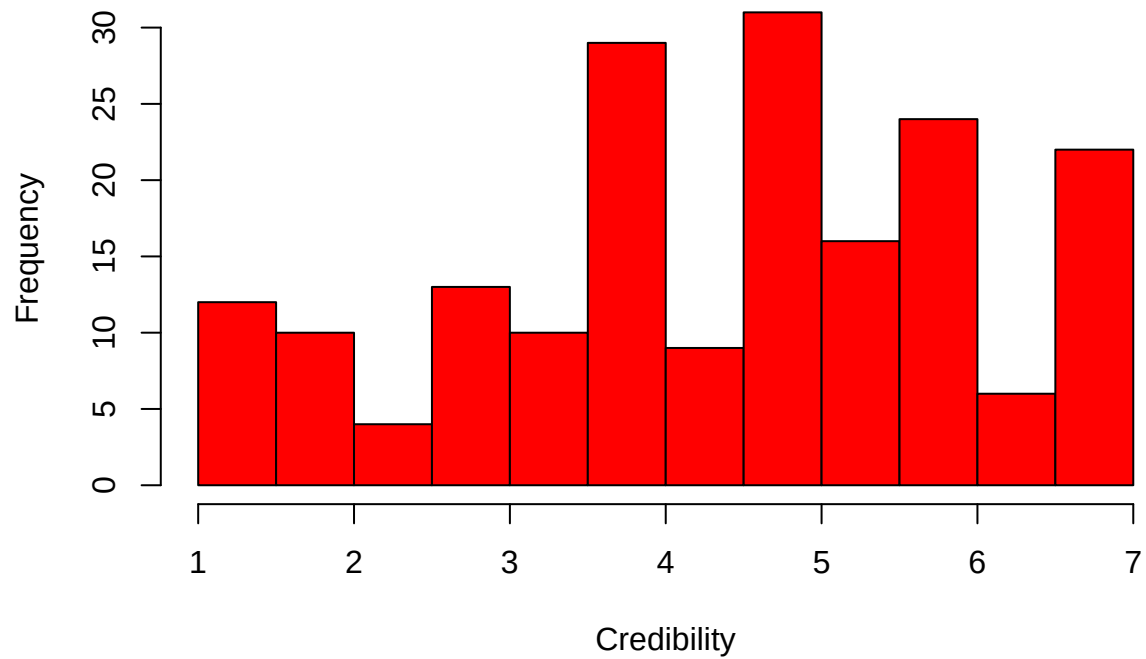
```
oil_means <- d[, .(Expert = mean(oil_exp_cred, na.rm = TRUE),  
                  Influencer = mean(oil_infl_cred, na.rm = TRUE))]  
barplot(as.numeric(oil_means), names.arg = c("Expert", "Influencer"),  
        main = "Average Oil Credibility", ylab = "Mean Credibility", col = "yellow")
```

Average Oil Credibility



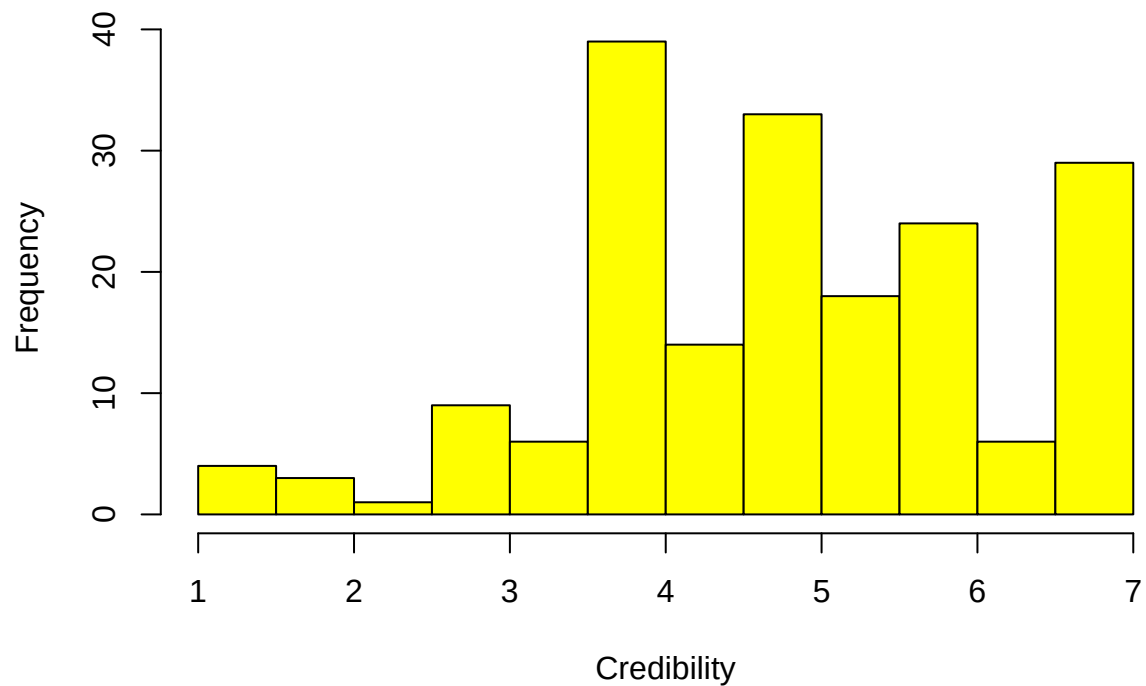
```
hist(d$meat_cred, main = "Distribution of Meat Credibility", xlab = "Credibility", col = "red")
```

Distribution of Meat Credibility



```
hist(d$oil_cred, main = "Distribution of Oil Credibility", xlab = "Credibility", col = "yellow")
```

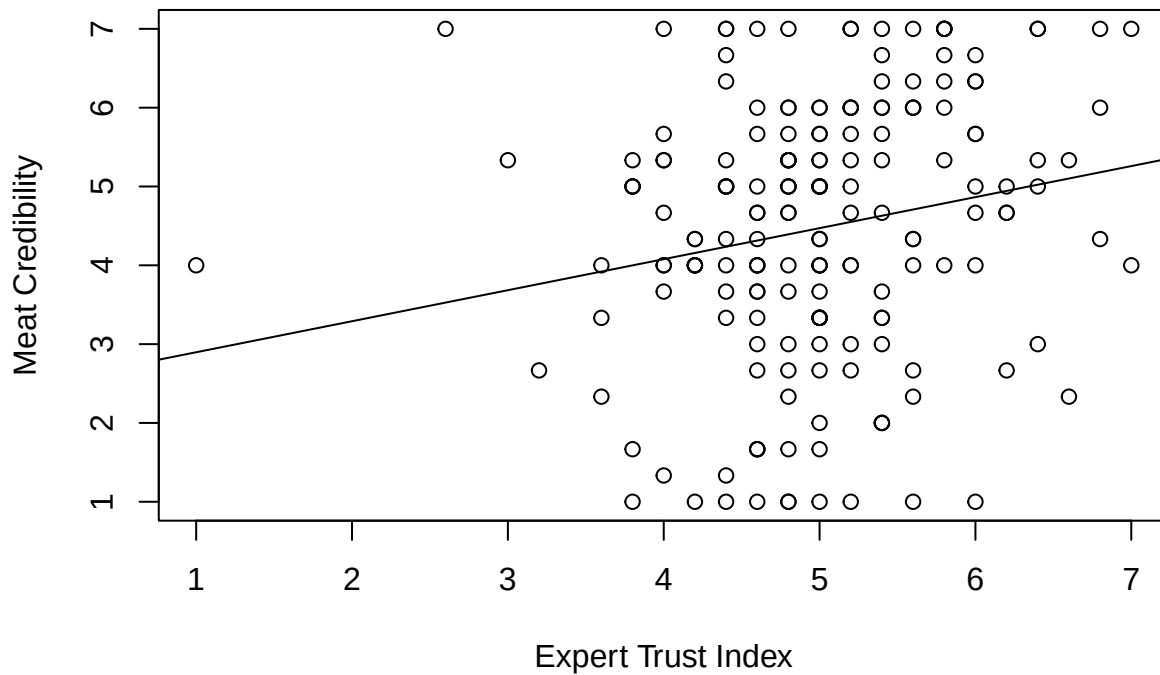
Distribution of Oil Credibility



```
plot(d$expert_trust_index, d$meat_cred,  
     main = "Trust in Experts vs Meat Credibility",  
     xlab = "Expert Trust Index", ylab = "Meat Credibility")
```

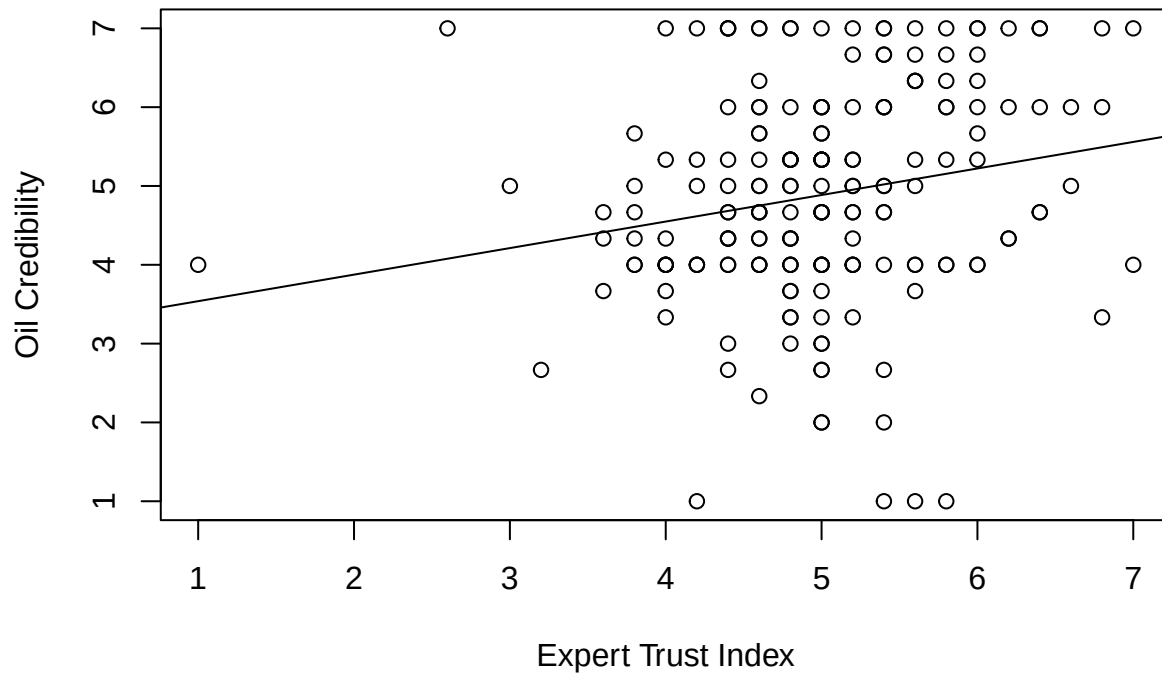
```
abline(lm(meat_cred ~ expert_trust_index, data = d))
```

Trust in Experts vs Meat Credibility



```
plot(d$expert_trust_index, d$oil_cred,  
     main = "Trust in Experts vs Oil Credibility",  
     xlab = "Expert Trust Index", ylab = "Oil Credibility")  
abline(lm(oil_cred ~ expert_trust_index, data = d))
```

Trust in Experts vs Oil Credibility

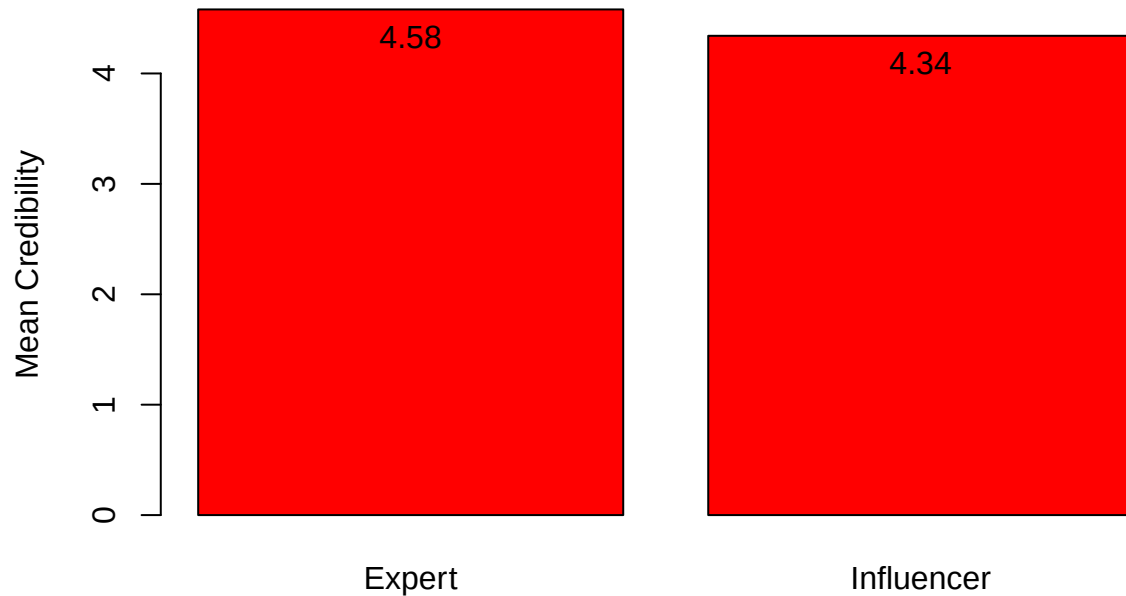


```
d2 <- d[SC0 == 0, ]

meat_means2 <- d2[, .(Expert      = mean(meat_exp_cred, na.rm = TRUE),
                    Influencer = mean(meat_infl_cred, na.rm = TRUE))]

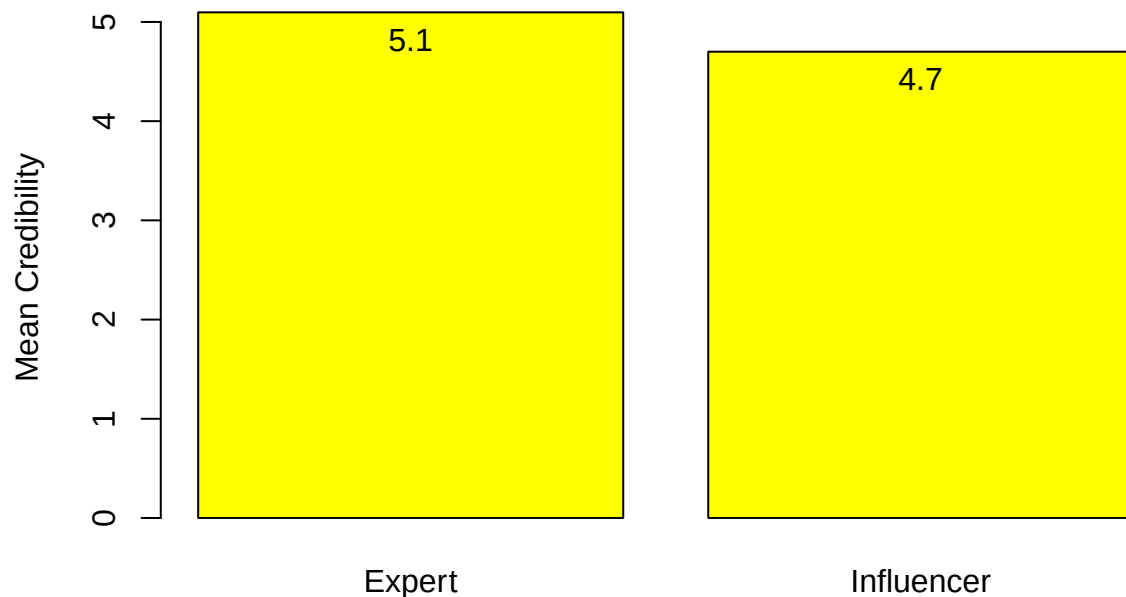
bp_meat <- barplot(as.numeric(meat_means2), names.arg = c("Expert", "Influencer"),
  main = "Average Meat Credibility", ylab = "Mean Credibility", col = "red")
text(x = bp_meat, y = as.numeric(meat_means2),
  labels = round(as.numeric(meat_means2), 2), pos = 1)
```

Average Meat Credibility



```
oil_means2 <- d2[, .(Expert = mean(oil_exp_cred, na.rm = TRUE),
                                Influencer = mean(oil_infl_cred, na.rm = TRUE))]
bp_oil <- barplot(as.numeric(oil_means2), names.arg = c("Expert", "Influencer"),
  main = "Average Oil Credibility", ylab = "Mean Credibility", col = "yellow")
text(x = bp_oil, y = as.numeric(oil_means2),
  labels = round(as.numeric(oil_means2), 2), pos = 1)
```

Average Oil Credibility



```
vals <- matrix(c(
  round(as.vector(meat_means2)$Expert, 2),
  round(as.vector(meat_means2)$Influencer, 2),
```

```

round(as.vector(oil_means2)$Expert, 2),
round(as.vector(oil_means2)$Influencer, 2)
), nrow = 2, byrow = FALSE)

rownames(vals) <- c("Expert", "Influencer")
colnames(vals) <- c("Meat", "Oil")

par(mar = c(6, 4, 4, 2))
bp <- barplot(vals, beside = TRUE,
  main = "Credibility by Source and Condition",
  ylab = "Mean Credibility",
  col = c("#4E79A7", "#F28E2B"))
text(x = bp, y = vals, labels = vals, pos = 1)
legend("bottom", inset = -0.25,
  legend = rownames(vals),
  fill = c("#4E79A7", "#F28E2B"),
  horiz = TRUE, bty = "n")

```

